YIDING MAO

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EDUCATION

University of Nottingham Ningbo China (2+2) Ningbo, China
Bachelor of Science in Statistics Sep 2023 to Present
University of Nottingham Nottingham, UK
Bachelor of Science in Statistics Sep 2025 to Present
Cumulative GPA: 3.98/4.00
Relevant Courses: *Differential Equations, Advanced Probability and Statistics, Mathematical Modeling, Markov chain, Coursera Stanford Online Machine Learning Specialization*

RESEARCH AND PROJECT EXPERIENCE

World Model for Autonomous Systems Ningbo, China
Supervised by Dr. Huan Jin in AIOP Research Group *Jun 2026 to Present*

- Reproduced a Conditional Diffusion Transformer (CDiT)-based Navigation World Model for autoregressive prediction of future visual observations conditioned on historical observations and actions.
- Deployed the **NoMaD (Goal Masked Diffusion Policy) model** with dynamic Loss Scaling to resolve the imbalance between Action and Distance Loss; designed an early stopping mechanism by introducing **cosine similarity** and achieved 0.95+ in cosine similarity of trajectory direction.
- Performed **real-world data acquisition** to further optimize the dataset by obtaining **relative position coordinates** and **trajectory yaw angles** for spatial-temporal pose labels, converting 80GB raw video stream into 20,000+ frames.
- Conducted a controlled ablation across training duration (**5-epoch** vs. **10-epoch** runs, **15 checkpoints** total) and identified that extended training improved trajectory-direction cosine similarity from ~**0.94 to 0.97+** while substantially reducing both GC distance loss and UC action loss, and further demonstrated — via independent random weight initialization — that checkpoints at equivalent epoch indices across separate training runs converge to distinct local optima rather than following comparable trajectories.

A Deep Reinforcement Learning - based Strategy for Instant Retail Supply Chain Interruption Ningbo, China
Supervised by Dr. Ruijin Wu *Dec 2025 to Present*

- Built a discrete-event simulation environment in Python with 94-dimensional **MDP state space** and 18 **action space** to simulate supply chain interruptions in a 1 Central Warehouse (CW), 10 Forward Warehouses (FWs), and 5 products environment.
- Implemented and compared **DQN, DDQN, Dueling DQN, D3QN, and PPO** in a risk-enhanced initial-state distribution to increase the exposure to severe disruption scenarios within the constrained training budget.
- Introduced **Min-Max and Standard normalization** for state preprocessing, and implemented an **Action Masking** mechanism to suppress invalid expediting actions during complete transportation failures; employed **Common Random Numbers (CRN)** across experimental conditions to significantly reduce evaluation noise introduced by Poisson demand and stochastic lead times.
- Evaluated learned policies against 4 **non-RL** baselines across 10 disruption scenarios using **Common Random Numbers**; **D3QN**, **Dueling DQN**, and **PPO** matched the strongest heuristic in profit, outperformed aggressive hedging by **5.5%**, and eliminated unnecessary expediting**.**

Clinical Mechanical Ventilation Modeling Ningbo, China
Supervised by Dr. Mainul Haque *Jul 2025 to Aug 2025*

- Developed mechanical ventilation models based on **ordinary differential equations (ODEs)** capturing dynamic changes in airway pressure, intrinsic PEEP, arterial blood gas levels, etc.; performed biological modeling and parameter optimization.
- Implemented **numerical simulations and visualizations** using **Python (NumPy) and MATLAB**; replicated and validated literature findings on the effect of PEEP on lung function in chronic airflow obstruction (CAO) patients; completed parameter optimization and result analysis.
- Successfully simulated airway pressure variations and gas exchange efficiency under different PEEP settings, providing evidence for individualized ventilation parameter adjustments in clinical settings.

PROFESSIONAL EXPERIENCE

Zhejiang Humanoid Robot Co., Ltd. Ningbo, China
Platform Software Intern *Jun 2026 to Sep 2026*

- Analyzed and validated the company's robot data platform across **Apache Pulsar**, **IoTDB**, **Doris**, **MySQL** and **Kubernetes**, tracing multi-source telemetry from real-time ingestion and messaging to time-series storage and analytical querying across **10+ telemetry and event streams**.
- Built data-quality validation workflows by reconciling timestamps and cross-source identifiers and detecting missing, duplicated, and inconsistent telemetry, achieving **95%+ cross-source record matching coverage** for downstream analytics and system debugging.
- Developed an **enterprise AI code-review system** based on Alibaba Open-Code-Review, integrating Git/CI, static analysis, repository-level retrieval, and LLM agents; evaluated against 200+ labeled defects, achieving approximately **31.9% precision**, **20.4% recall**, and **0.18 F1**.
- Optimized diff-aware context retrieval and LLM review workflows, increasing true-positive defect detection by **7.7%** on representative code changes.

BOE Technology Group Co. Beijing, China
Data Analyst Intern *Jun 2025 to Aug 2025*

- Developed an employee-attrition analytics framework using **50,000+ records from 2022–2025**, integrating personnel, factory, operational, and external benchmark data to support workforce-risk planning.
- Cleaned and standardized multi-source records handled missing and anomalous values, and engineered **20+ employee- and department-level features** including overtime intensity, compensation deviation, tenure, and historical turnover indicators.
- Built and compared linear regression, logistic-regression, and decision-tree models for department-level turnover forecasting and individual attrition prediction, achieving **0.84 AUC and 78% recall** on the best-performing classifier.
- Translated model outputs into actionable workforce insights through **Power BI** dashboards and presented key attrition drivers and retention recommendations to departmental stakeholders.

Donghai Insurance Ningbo, China
Data Analyst Intern *Jul 2024 to Aug 2024*

- Built a **Python-based** data preprocessing workflow to clean and structure **20,000+** policy and claims records, constructing policyholder- and policy-level features for pricing analysis.
- Developed frequency and severity models to decompose expected loss — Poisson regression with a **log-exposure offset** to model annualized claim frequency, and a **log-normal** severity model fitted on log-transformed claim amounts over the claims-incurred subset — using **15+** policyholder and policy features; decision trees were fitted as a benchmark and to surface candidate interactions.
- Combined predicted frequency and severity to estimate expected loss and evaluate loss ratios across risk segments, supporting tiered coverage and premium-pricing recommendations.
- Validated model stability with **5-fold cross-validation** and used feature-importance analysis to identify the dominant risk factors, which were translated into risk-tier and premium-rating recommendations.

SKILLS

Programming Languages: Python, R, SQL
Python Expertise: Sklearn, TensorFlow, OpenCV, PyTorch, Numpy
Processing Software: Unity Engine, Adobe Premire
Visualization: Tableau, Power BI, Flask, HTML